From Predictive Uncertainty to Decision-Effective Dynamics Ambiguity Sets

Perturbation-Calibrated Function-Space Uncertainty for Robust PDE Control

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Abstract

Neural PDE surrogates are usually assessed by prediction error, whereas a controller is affected by the component of model error that changes its finite-horizon objective. This paper studies the interface between these two objects: how can predictive uncertainty be converted into a dynamics ambiguity set that is effective for control decisions? We train a residual Fourier neural operator (FNO) and a replica on Gaussian-perturbed labels. Their smoothed disagreement supplies a spatial scale, while a disjoint deployment audit split supplies finite-sample calibration. The resulting normalized L^2 ellipsoid and simultaneous coordinate box are field-valued dynamics sets, not scalar error bars. Robust model predictive control (MPC) queries their geometry either through exact support functions in the finite-horizon adjoint direction or through projected-gradient adversarial rollouts. We prove split-conformal marginal coverage on a fixed discretization, derive exact ellipsoidal and box support functions and a second-order linearization remainder, and establish multi-step rollout and discounted policy-transfer bounds under a separate uniform one-step error assumption. The conformal and uniform-error results are explicitly not merged. On controlled viscous Burgers dynamics under parameter, boundary, initial-condition, and latent actuator shifts, deployment calibration restores field coverage and changes closed-loop tail cost in a geometry-dependent manner; the simultaneous-box adjoint controller lowers empirical $p90$ cost by 10.84% relative to nominal MPC in 24 matched cases. An official two-dimensional Navier–Stokes benchmark exposes the high-dimensional cost of simultaneous bands and weak error localization. These results support an uncertainty-to-decision mechanism, not a general safety certificate.

1 Introduction

Learned operators can replace repeated numerical solves inside planning and control loops. Fourier neural operators (FNOs), DeepONets, and related neural operators learn maps between discretized functions and have demonstrated fast surrogate prediction for Burgers, Darcy, and Navier–Stokes systems [13] and [14]; broader operator-learning foundations are developed in [11] and [4]. Their speed addresses only one part of model-based control. Once a controller optimizes actions through a learned propagator, it can amplify a small but decision-aligned model error even when the average field error is low.

Predictive uncertainty quantification (UQ) does not by itself resolve this problem. In a spatial field, a large residual in a dynamically inactive region may barely change the cost, while a smaller residual near an actuator or a high-value gradient can reverse the preferred action. Conformal prediction (CP) can give distribution-free finite-sample coverage for a chosen score under exchangeability [1]; see also the foundational treatment in [26]. However, coverage specifies an event, not the action that should be taken. Conversely, robust MPC requires an uncertainty geometry and repeatedly queries its support along objective-sensitive directions. The scientific gap is therefore not "how to attach error bars to an FNO", but how to turn a calibrated field-valued prediction set into a control object without overstating what the coverage theorem guarantees.

We ask:

> *How can predictive uncertainty be converted into a dynamics ambiguity set that is effective for control decisions?*

Our answer is a three-stage construction: learn a local scale from sensitivity to label perturbations; calibrate that scale on a disjoint deployment audit; and query the resulting function-space set through the robust MPC objective. The FNO is a replaceable world model rather than the architectural contribution.

Table 1: Positioning relative to the closest lines of work. "Field" means a joint object over a discretized PDE output; guarantee scopes are not directly comparable.

Work	Uncertainty object	Decision interface	Main guarantee scope
Ma et al. [15]	Functional quantile region	None	Risk-controlled functional coverage
Wang et al. [27]	Perturbation-scaled field box	None	Simultaneous discretized-field coverage
Chee et al. [6]	Weighted-CP state intervals	Constraint tightening and reference generation	Time-varying model accuracy and robust constraints
Srinivasan et al. [23]	Covariance-shaped OOD tube	System-level robust MPC	Weighted-CP OOD coverage and tube robustness
This work	Audit-calibrated L^2 ball, ellipsoid, and field box	Adjoint support or nonlinear adversarial MPC	One-step/behavior-rollout CP; separate deterministic value bound

1.1 Related work

Operator learning and uncertainty. FNOs parameterize global integral kernels in Fourier space [13], while the broader neural-operator framework formalizes discretization-invariant maps between function spaces [11]. DeepONet provides a complementary branch–trunk construction [14]. Recent UQ work includes deep ensembles under PDE shift [20], probabilistic neural operators [5], approximate Bayesian neural operators [16], and linearized function-valued Gaussian processes [17]. These methods estimate structured uncertainty, but Bayesian or ensemble dispersion is not automatically a finite-sample coverage statement.

Conformal operator-learning methods address validity in different senses. Risk-controlling quantile neural operators calibrate functional coverage [15]; Conformalized-DeepONet wraps a DeepONet with distribution-free bands [21]; and split CP has been formulated directly in function space [19]. Most closely, Wang et al. train clean-label and perturbed-label FNOs and calibrate a max-type score for simultaneous Navier–Stokes fields in a data-scarce regime [27]. We adopt the perturbation scale as a sample-efficient starting point, then ask the downstream question left open by field coverage: which calibrated geometry is useful for control?

Conformal prediction for robust decisions and control. Conformal uncertainty sets have been connected to robust optimization [10]. In control, Chee et al. combine receding-horizon weighted CP, dynamic constraint tightening, and a predictive reference generator [6]; adaptive CP has also been used in motion planning [8]. SODA-MPC uses conformalized ensembles and runtime OOD adaptation [7], while conformalized system-level synthesis constructs OOD reachable tubes [23]. Those works target finite-dimensional state tubes, constraints, or online drift. Our setting instead starts with a discretized PDE field, compares multiple function-space geometries at a matched audit budget, and uses the finite-horizon objective itself to query the set. We do not inherit their sequential, feasibility, or safety guarantees.

Decision-aware model error. Value-aware model learning replaces uniform prediction quality with error measured through downstream value [9]; calibrated model-based reinforcement learning likewise shows that uncertainty quality can affect planning [18]. Simulation lemmas and Lipschitz analyses propagate one-step model error to long-horizon value error [2]. Distributionally robust model-based reinforcement learning optimizes against dynamics ambiguity in large state spaces [24]. Our contribution is an explicit bridge among these themes: conformal calibration constructs the field set, its support function connects geometry to value sensitivity, and independent closed-loop tests decide whether the construction is useful.

1.2 Contributions

The paper makes four scoped contributions.

(i) It defines a reproducible data pipeline from proper training through a disjoint deployment audit to function-space dynamics sets. A clean-label FNO and a perturbed-label replica use the full proper-training input set; no calibration label trains either predictor.

(ii) It converts the sets into decision objects. We derive exact support functions for normalized L^2 ellipsoids and coordinate boxes, use them in a finite-horizon adjoint robust counterpart, and compare that approximation with nonlinear projected-gradient adversarial rollouts.

(iii) It separates three kinds of theory: finite-sample marginal conformal coverage, deterministic multi-step/value propagation under a uniform error assumption, and a conditional curvature remainder for the first-order robust objective. No one of these statements is used as a substitute for another.

(iv) It evaluates coverage, tightness, and control independently. Controlled Burgers supplies closed-loop evidence under compound shift; the official Navier–Stokes data supply a high-dimensional field-UQ stress test but no control claim. Negative results are retained.

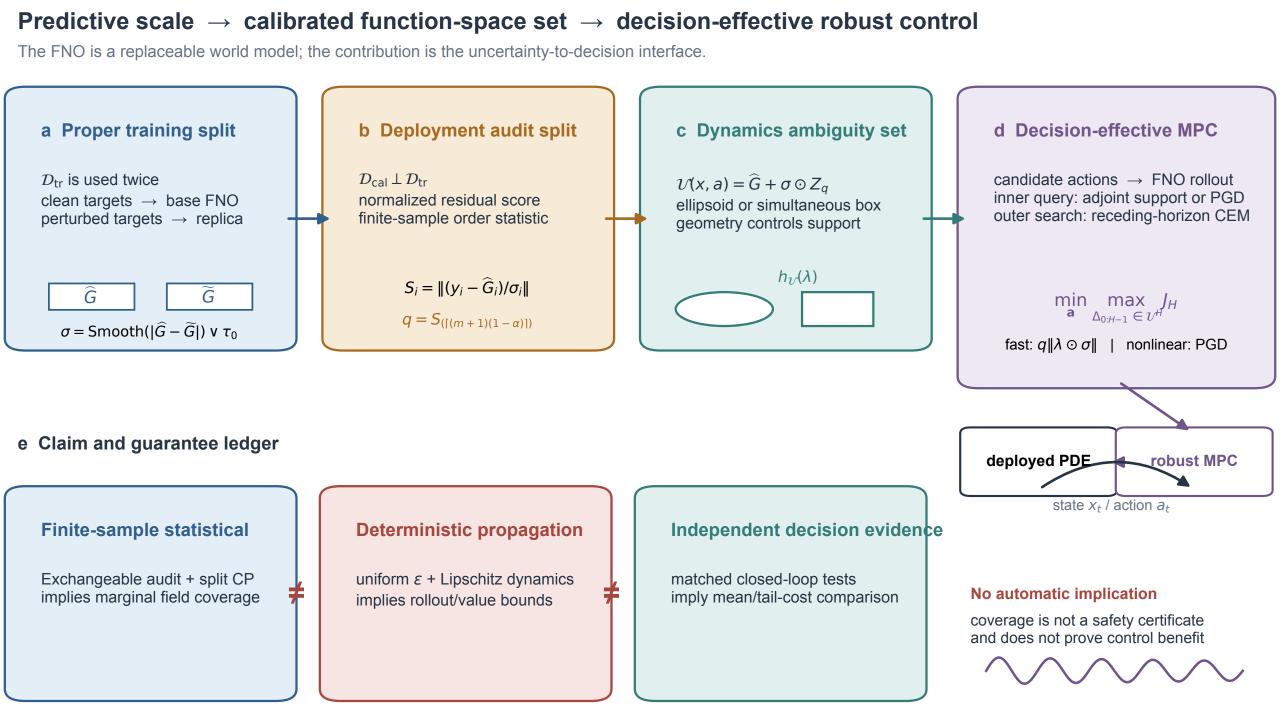

Figure 1: Schematic of the proposed uncertainty-to-decision pipeline. Proper training, deployment calibration, and independent evaluation use disjoint data roles. The ambiguity set enters robust MPC through its support function or a nonlinear adversarial rollout. The lower ledger prevents three statements from being conflated: split-conformal field coverage, deterministic uniform-error propagation, and empirical closed-loop benefit.

2 Preliminaries and problem setting

2.1 Controlled function-valued dynamics and discretization

Let $u_t \in \mathcal{H}$ be a field on a spatial domain Ω, let $a_t \in \mathcal{A} \subset \mathbb{R}^p$ be a control, and let ξ collect physical parameters, boundary conditions, forcing, and unobserved actuator properties. The deployment transition is

$$u_{t+1} = \mathcal{G}_\star(u_t, a_t; \xi). \tag{1}$$

For computation, a projection $P_n : \mathcal{H} \to \mathbb{R}^n$ produces $x_t = P_n u_t$. We write

$$x_{t+1} = G_{\star,n}(x_t, a_t; \xi), \qquad \langle v, w \rangle_n = \frac{1}{n} \sum_{j=1}^{n} v_j w_j, \qquad \|v\|_{2,n}^2 = \langle v, v \rangle_n \,. \tag{2}$$

Nonuniform quadrature weights can replace $1/n$ without changing the construction. Throughout, "function-space set" means a joint set for the entire discretized field. Coverage on one fixed grid is not, by itself, a continuum or grid-transfer guarantee.

The finite-horizon control cost is

$$J_H(x_0, a_{0:H-1}) = \sum_{t=1}^{H} \ell(x_t) + \sum_{t=0}^{H-1} r(a_t), \tag{3}$$

where low cost is preferred. A learned one-step propagator $\widehat{G}_\theta$ is rolled out inside MPC and can therefore be queried at states and actions induced by the optimizer rather than by the data-collection policy.

2.2 Data roles and deployment shift

We distinguish four data roles:

$$\mathcal{D}_{\mathrm{tr}} = \{(x_i, a_i, y_i)\}_{i=1}^{N_{\mathrm{tr}}}, \qquad \mathcal{D}_{\mathrm{val}} = \{(x_i, a_i, y_i)\}_{i=1}^{N_{\mathrm{val}}},$$
$$\mathcal{D}_{\mathrm{cal}} = \{(x_i, a_i, y_i)\}_{i=1}^{m}, \qquad \mathcal{D}_{\mathrm{te}} = \{(x_i, a_i, y_i)\}_{i=1}^{N_{\mathrm{te}}}.$$

Proper training and validation come from a source regime. The deployment audit $\mathcal{D}_{\mathrm{cal}}$ is a small labeled sample from the target regime and is used only to compute conformal order statistics. The test set remains untouched until final evaluation. The clean and perturbed models, smoothing kernel, and floor rule are measurable with respect to proper-training randomness and are fixed before $\mathcal{D}_{\mathrm{cal}}$ is scored.

This is an *audit-calibrated* setting: validity requires that calibration and target examples are exchangeable after conditioning on the trained models. If the deployment distribution changes again after the audit, ordinary split CP does not protect against that new shift. Weighted or adaptive CP [3], or covariate-shift calibration [25], is a possible extension, not a guarantee claimed here.

2.3 Split conformal prediction

Let $Z_i = (X_i, A_i, Y_i)$ and let $S(Z_i)$ be a fixed scalar nonconformity score. With calibration scores $S_{(1)} \le \cdots \le S_{(m)}$, target miscoverage $\alpha \in (0, 1)$, and

$$k = \lceil (m+1)(1-\alpha) \rceil, \qquad q_{1-\alpha} = S_{(k)}, \tag{4}$$

the standard convention sets $q_{1-\alpha} = +\infty$ if $k > m$. Equivalently, the empirical distribution is augmented by an atom at $+\infty$. For a new input (x, a), the prediction set is

$$C(x, a) = \{y : S(x, a, y) \le q_{1-\alpha}\}. \tag{5}$$

Under exchangeability, the rank of the new score among the $m+1$ scores yields marginal coverage at least $1-\alpha$. The word *marginal* matters: it is not conditional coverage at every state–action pair and it is not automatic simultaneous coverage over a counterfactual MPC rollout.

2.4 Problem statement and decision effectiveness

We seek a map from predictive scale to a dynamics set $\mathcal{U}_\theta(x, a) \subset \mathbb{R}^n$, followed by a robust planning rule. A useful construction must answer two different questions:

(P1) **Predictive validity and efficiency:** does the set attain its predeclared coverage event on independent deployment data without unnecessary width?

(P2) **Decision effectiveness:** at a matched world model and audit budget, does using the set reduce paired mean cost, upper-tail cost, or constraint violation relative to nominal planning and isotropic-set baselines?

Definition 1 (Decision-effective ambiguity set)**.** *For a predeclared deployment distribution, controller, and downstream metric, an ambiguity-set construction is decision-effective if its robust controller improves that metric on independent matched closed-loop cases without violating the accompanying mean-cost or feasibility criterion. Coverage is supporting evidence, not the definition.*

3 Methodology

3.1 Residual FNO world model

The base one-step world model is a residual FNO,

$$\widehat{G}_\theta(x, a, \xi_{\mathrm{obs}}) = x + \delta_\theta(x, a, \xi_{\mathrm{obs}}). \tag{6}$$

For Burgers, the five input channels are the current field, the action multiplied by a fixed actuator profile, viscosity, a linear extension of the two boundary values, and the spatial coordinate. Each block applies a spectral convolution and a pointwise linear map inside a ResNet-style skip connection. The final boundary coordinates are projected exactly to the supplied Dirichlet values. The full run uses width 32, 14 Fourier modes, four residual blocks, and 240,454 trainable parameters across the clean and perturbed operators.

FNO is chosen because it is a standard field propagator [13]; the broader framework is reviewed in [11]. Nothing in the calibration or robust-control construction requires Fourier layers. Replacing it by another differentiable operator changes the mean and scale quality, not the logic of the method.

3.2 Perturbation-based uncertainty scale

Following the stability principle used for data-scarce operator UQ [27], we train a second operator with the same architecture and the same proper-training inputs but perturbed labels:

$$\widetilde{y}_i = y_i + \eta_i, \qquad \eta_i \overset{\text{iid}}{\sim} \mathcal{N}(0, c^2 s_y^2 I_n). \tag{7}$$

Let $\widetilde{G}_{\widetilde{\theta}}$ be its prediction. The raw disagreement, smoothed disagreement, and stabilized scale are

$$d(x,a) = \left| \widehat{G}_\theta(x,a) - \widetilde{G}_{\widetilde{\theta}}(x,a) \right|, \tag{8}$$

$$\bar{d}(x,a) = K * d(x,a), \tag{9}$$

$$\sigma(x,a) = \bar{d}(x,a) \vee \tau_0. \tag{10}$$

The convolution is an odd moving-average window (five coordinates for Burgers; 15×15 for Navier–Stokes). On Burgers, τ_0 is 10% of the median interior disagreement on $\mathcal{D}_{\text{tr}}$, lower bounded by 10^{-6}. Smoothing prevents isolated near-zero denominators from dominating a max-type score; the floor is fixed before calibration. The raw scale has no coverage interpretation. Split conformal calibration supplies validity.

3.3 Perturbation-based nonconformity scores and calibration

For residual $r = y - \widehat{G}_\theta(x,a)$, we compare three geometries.

$$S_{\text{L2}}(x,a,y) = \frac{\|r\|_{2,n}}{\|\sigma(x,a)\|_{2,n}}, \tag{11}$$

$$S_2(x,a,y) = \|r \oslash \sigma(x,a)\|_{2,n}\,, \tag{12}$$

$$S_\infty(x,a,y) = \|r \oslash \sigma(x,a)\|_\infty\,. \tag{13}$$

Here $\oslash$ denotes coordinate-wise division. Applying Equation (4) separately gives q_{L2}, q_2, and q_∞. The corresponding one-step dynamics sets are

$$\mathcal{U}_{\text{L2}}(x,a) = \left\{ \widehat{G}_\theta(x,a) + \Delta : \|\Delta\|_{2,n} \le q_{\text{L2}}\, \|\sigma(x,a)\|_{2,n} \right\}, \tag{14}$$

$$\mathcal{U}_2(x,a) = \left\{ \widehat{G}_\theta(x,a) + \Delta : \|\Delta \oslash \sigma(x,a)\|_{2,n} \le q_2 \right\}, \tag{15}$$

$$\mathcal{U}_\infty(x,a) = \left\{ \widehat{G}_\theta(x,a) + \Delta : |\Delta_j| \le q_\infty \sigma_j(x,a),\ j = 1, \ldots, n \right\}. \tag{16}$$

The first is an isotropic L^2 tube baseline. The second preserves heteroscedastic anisotropy but covers an L^2 event. The third targets simultaneous containment of every output coordinate. The box is usually wider because one failure anywhere in the field breaks the event.

3.4 Practical implementation

The complete calibration procedure is:

Step 1. Train $\widehat{G}_\theta$ on clean proper-training targets and $\widetilde{G}_{\widetilde{\theta}}$ on perturbed targets; select checkpoints using only $\mathcal{D}_{\text{val}}$.

Step 2. Freeze both operators. Compute the smoothing rule and τ_0 from proper-training predictions.

Step 3. On each calibration triple in $\mathcal{D}_{\text{cal}}$, compute the mean, disagreement scale, residual, and the three scores in Equations (11) to (13).

Step 4. Compute the finite-sample order statistics in Equation (4). Store the world-model weights, scale hyperparameters, calibration multipliers, data-split metadata, and random seed.

Step 5. At deployment, evaluate $\widehat{G}_\theta$ and σ for each MPC query and pass the selected set geometry to the robust inner problem. No calibration label is reused for final coverage or control evaluation.

4 Decision-effective robust MPC

4.1 Rectangular robust planning problem

For a candidate action sequence, nominal MPC minimizes $\widehat{J}_H$ obtained by recursively applying $\widehat{G}_\theta$. We use a rectangular product of one-step sets as a robust planning model:

$$\min_{a_{0:H-1}\in\mathcal{A}^H}\ \max_{\Delta_t\in\mathcal{E}_t(a_{0:t})} \sum_{t=1}^{H}\ell(x_t)+\sum_{t=0}^{H-1}r(a_t), \quad x_{t+1}=\widehat{G}_\theta(x_t,a_t)+\Delta_t, \tag{17}$$

where $\mathcal{E}_t$ is the centered version of either Equation (15) or Equation (16), evaluated along the perturbed rollout. Rectangularity makes the inner search tractable but is a modeling choice, not a consequence of marginal CP coverage.

4.2 Exact support and the adjoint robust counterpart

Let λ_{t+1} denote the full derivative of the remaining nominal horizon cost with respect to x_{t+1}, expressed in the normalized inner product: $\delta J = \langle\lambda_{t+1},\Delta_t\rangle_n$. Its value includes the effect of Δ_t on every later predicted state. The support of a centered set $\mathcal{E}$ is $h_{\mathcal{E}}(\lambda)=\sup_{\Delta\in\mathcal{E}}\langle\lambda,\Delta\rangle_n$.

Proposition 1 (Support functions of the calibrated geometries)**.** *For positive scale σ,*

$$h_{\mathcal{E}_2}(\lambda)=q_2\left\|\lambda\odot\sigma\right\|_{2,n}, \tag{18}$$

$$h_{\mathcal{E}_\infty}(\lambda)=\frac{q_\infty}{n}\sum_{j=1}^{n}|\lambda_j|\sigma_j. \tag{19}$$

Proof. For the ellipsoid, set $\Delta=\sigma\odot z$ and apply Cauchy–Schwarz under $\langle\cdot,\cdot\rangle_n$; equality is attained by choosing z parallel to $\lambda\odot\sigma$. For the box, every coordinate is independent, and the maximizer is $\Delta_j=q_\infty\sigma_j\,\mathrm{sign}(\lambda_j)$. ☐

Linearizing the stacked inner problem at the nominal rollout gives

$$\widehat{J}_H(a_{0:H-1})+\sum_{t=0}^{H-1}h_{\mathcal{E}_t}(\lambda_{t+1}). \tag{20}$$

This formula is the decision interface: two sets can have similar coverage but produce different robust costs because their support differs along λ_{t+1}. Automatic differentiation supplies all finite-horizon adjoints in one reverse pass.

4.3 Nonlinear adversarial rollout

The first-order objective can miss state-dependent scales and nonlinear error propagation. Our second inner solver parameterizes $\Delta_t=\sigma(x_t,a_t)\odot z_t$, initializes $z_{0:H-1}=0$, performs projected gradient ascent on the full rollout cost, and projects each z_t onto either $\|z_t\|_{2,n}\le q_2$ or $\|z_t\|_\infty\le q_\infty$. The experiment uses three ascent steps. This is an approximate nonconvex inner maximization, not an optimality certificate.

4.4 Outer optimization and receding-horizon deployment

The outer action sequence is optimized by the cross-entropy method (CEM): 128 candidates, 16 elites, four iterations, horizon eight, and action bound $|a_t|\le 2$. The first action is applied to the numerical plant and the entire problem is replanned at the next observed state. Nominal, isotropic-tube, adjoint-ellipsoid, adjoint-box, and adversarial controllers use the same world model and CEM budget.

5 Theoretical analysis

5.1 Finite-sample coverage on a fixed discretization

Theorem 1 (One-step split-conformal coverage)**.** *Condition on $\mathcal{D}_{\mathrm{tr}}$, model-training randomness, perturbation noise, and all scale hyperparameters. If the m calibration triples and one new deployment triple are exchangeable, then each set constructed by its matching score satisfies*

$$\Pr\{G_{\star,n}(X,A)\in\mathcal{U}(X,A)\}\ge 1-\alpha.$$

For $\mathcal{U}_\infty$, the event contains all n output coordinates simultaneously.

Proof. After conditioning, the score is fixed and identical for every example. The m calibration scores and the new score are therefore exchangeable. The rank argument for Equation (4), including the $+\infty$ convention, gives the result. For the max score, $S_\infty \le q_\infty$ is equivalent to all n coordinate inequalities in Equation (16). □

The theorem concerns one random transition. To obtain a different, valid trajectory statement, predeclare a behavior policy and define on whole length-H trajectories

$$S^{\text{traj}} = \max_{0 \le t < H} \max_{1 \le j \le n} \frac{|x_{t+1,j} - \widehat{x}_{t+1,j}|}{\sigma_j(\widehat{x}_t, a_t)}. \tag{21}$$

Split-conformal calibration of this scalar score on exchangeable trajectories covers all times and coordinates of one new behavior-policy trajectory with probability at least $1-\alpha$. It does not certify counterfactual action sequences chosen by MPC.

5.2 First-order robustification and curvature

Let $\boldsymbol{\Delta} = (\Delta_0, \ldots, \Delta_{H-1})$ and let $F(\boldsymbol{\Delta})$ be the finite-horizon cost after injecting the stacked perturbation. Let $\mathcal{E}^H$ be the product of the centered one-step sets.

Proposition 2 (Conditional second-order remainder)**.** *Suppose F is twice differentiable and $\|\nabla^2 F(\boldsymbol{z})\|_{\text{op}} \le M_H$ on the line segments joining 0 to every $\boldsymbol{\Delta} \in \mathcal{E}^H$. Then*

$$\sup_{\boldsymbol{\Delta} \in \mathcal{E}^H} F(\boldsymbol{\Delta}) \le F(0) + \sum_{t=0}^{H-1} h_{\mathcal{E}_t}(\lambda_{t+1}) + \frac{M_H}{2} \sup_{\boldsymbol{\Delta} \in \mathcal{E}^H} \|\boldsymbol{\Delta}\|_2^2. \tag{22}$$

Proof. Taylor's theorem gives $F(\boldsymbol{\Delta}) \le F(0) + \langle \nabla F(0), \boldsymbol{\Delta} \rangle + (M_H/2)\|\boldsymbol{\Delta}\|_2^2$. The support of a Cartesian product is the sum of the component supports. Taking the supremum gives Equation (22). □

The adjoint objective retains only the first two terms. A learned or audit-fitted quadratic feature is not a certified curvature bound unless a uniform M_H is established.

5.3 Multi-step propagation under uniform one-step error

Conformal coverage is distributional. The following result uses a different, deterministic assumption.

Assumption 1 (Uniform model error and Lipschitz dynamics)**.** *On a forward-invariant state–action region $\mathcal{K}$,*

$$\left\| G_{\star,n}(x,a) - \widehat{G}_\theta(x,a) \right\|_{2,n} \le \epsilon, \qquad \|G_{\star,n}(x,a) - G_{\star,n}(x',a)\|_{2,n} \le L_G \|x - x'\|_{2,n}. \tag{23}$$

Proposition 3 (Open-loop rollout error)**.** *For a common action sequence, equal initial states, and $e_t = \|x_t - \widehat{x}_t\|_{2,n}$,*

$$e_{t+1} \le L_G e_t + \epsilon, \qquad e_h \le \epsilon \sum_{k=0}^{h-1} L_G^k. \tag{24}$$

Proof. Add and subtract $G_{\star,n}(\widehat{x}_t, a_t)$, then use the two parts of Equation (23). Iterating the scalar recurrence with $e_0 = 0$ gives the geometric sum. □

5.4 Discounted value and policy transfer

For a feedback policy π, define the closed-loop map $F_G^\pi(x) = G(x, \pi(x))$ and stage reward $r_\pi(x) = r(x, \pi(x))$. The value is $V_G^\pi(x_0) = \sum_{t=0}^\infty \gamma^t r_\pi(x_t)$.

Theorem 2 (Fixed-policy value error)**.** *Suppose the true and learned closed-loop maps for a fixed policy share the state Lipschitz constant L_G, their one-step discrepancy is uniformly at most ϵ, r_π is L_r-Lipschitz, and $\gamma L_G < 1$. Then*

$$\left| V_{G_\star}^\pi(x_0) - V_{\widehat{G}}^\pi(x_0) \right| \le B(\epsilon) := \frac{\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)}. \tag{25}$$

Proof. By Equation (24), $e_t \le \epsilon \sum_{k=0}^{t-1} L_G^k$. Hence

$$\sum_{t=1}^{\infty} \gamma^t L_r e_t \le L_r \epsilon \sum_{k=0}^{\infty} L_G^k \sum_{t=k+1}^{\infty} \gamma^t = \frac{\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)}.$$

□

Corollary 1 (Policy transfer with optimization error)**.** *Suppose the preceding assumptions hold uniformly over a policy class. Let $\pi^\star$ maximize true-model value, and let $\widehat{\pi}$ satisfy*

$$\sup_{\pi} V_{\widehat{G}}^{\pi}(x_0) - V_{\widehat{G}}^{\widehat{\pi}}(x_0) \le \delta_{\text{opt}}. \tag{26}$$

Then

$$V_{G_\star}^{\pi^\star}(x_0) - V_{G_\star}^{\widehat{\pi}}(x_0) \le \frac{2\gamma L_r \epsilon}{(1-\gamma)(1-\gamma L_G)} + \delta_{\text{opt}}. \tag{27}$$

If $L_G \le 1$, the model term is at most $2\gamma L_r \epsilon/(1-\gamma)^2$.

Proof. Insert and subtract the two learned-model values. The two model-transfer terms are bounded by $B(\epsilon)$, and learned-model suboptimality contributes δ_{opt}. □

The CEM controller has finite sampling and four optimization iterations, so δ_{opt} is not known in our experiment. The corollary explains the target scaling; it is not a certificate for the reported CEM policy.

5.5 Guarantee separation

Table 2: The three statement types used in the paper. None implies either of the others without additional assumptions.

Statement	Assumption	Event or quantity	What it does not provide
Split CP	Exchangeable audit/test scores	Marginal one-step field containment	Uniform error over a region or MPC safety
Trajectory CP	Exchangeable behavior-policy trajectories	All time/coordinates for one such trajectory	Counterfactual MPC action coverage
Value bound	Deterministic uniform ϵ, Lipschitz maps/reward	Rollout and policy-transfer error	A probability statement or conformal validity
Closed-loop test	Independent matched cases	Empirical mean/tail cost	A theorem beyond the tested design

6 Experimental design

6.1 Research questions

The experiments answer five predeclared questions.

RQ1. Does deployment-audit calibration restore field coverage under compound shift relative to source calibration?

RQ2. At the same model and deployment audit, do ambiguity geometry and decision querying change mean or upper-tail closed-loop cost?

RQ3. Does the deterministic value-gap construction exhibit the predicted $O(\epsilon)$ and $O((1-\gamma)^{-2})$ scaling?

RQ4. How does the perturbation scale behave on a high-dimensional Navier–Stokes field, where a simultaneous max event is demanding?

RQ5. At matched independent-test coverage, are local or adjoint value bounds less conservative than a global recursion?

Table 3: Disjoint data roles in the final experiments. Burgers one-step test size is per regime. Trajectory calibration/test splits are separate from all one-step splits.

Benchmark	Train	Validation	Source cal.	Deploy. audit	Test	Traj. audit/test
Burgers	4000	600	800	300	800	200/400
Navier–Stokes	800	50	–	200	300	–

6.2 Controlled Burgers benchmark

The plant is the one-dimensional viscous Burgers equation

$$u_t + uu_x = \nu u_{xx} + g_{\mathrm{act}} a_t b(x), \quad x \in [0, 1], \quad u(t, 0) = b_L, quadu(t, 1) = b_R. \tag{28}$$

It is solved on 64 grid points with an upwind convective flux, centered diffusion, solver step 2×10^{-4}, and control interval 0.02. The Gaussian actuator is centered at $x = 0.68$. The stage cost is

$$\ell(u) + r(a) = \frac{1}{n} \sum_{j=1}^{n} w(x_j) u_j^2 + 0.002a^2, \quad w(x) = 0.15 + 0.85 \exp\left[-\frac{1}{2}\left(\frac{x - 0.72}{0.14}\right)^2\right]. \tag{29}$$

Training and deployment differ in viscosity, boundary values, initial-field amplitude, and latent actuator gain g_{act}. The model observes viscosity and boundaries but not the actuator gain.

6.3 Two-dimensional Navier–Stokes uncertainty benchmark

We additionally use the official NeuralOperator incompressible Navier–Stokes benchmark [22]. The downloaded Reynolds-500 archive supplies 128×128 vorticity input–output fields. A clean FNO and a perturbed-label FNO are trained on disjoint proper-training indices, followed by a max-score conformal audit. The dataset contains no action channel; it tests field UQ and dimensionality, not closed-loop control.

6.4 Baselines, metrics, and implementation

All Burgers controllers use the same residual FNO pair. We compare: uncontrolled dynamics; nominal MPC; source-calibrated isotropic L^2 robust MPC; deployment-audit L^2 robust MPC; audit ellipsoid with adjoint support; audit simultaneous box with adjoint support; and audit ellipsoid with nonlinear adversarial rollouts. This design separates access to deployment labels from ambiguity geometry and from the inner solver.

We report realized function-set coverage, decision-linear coverage (residual projected onto a cost direction), mean support, radius–error association, behavior-trajectory coverage, paired mean control cost, empirical $p90$ cost, and mean absolute action. Twenty-four combined-shift actuator-gain cases are matched across controllers, each with 20 receding-horizon steps. These cases share the same initial field and physical parameters except actuator gain; they are a controlled sweep rather than 24 independent PDE draws.

The two FNOs are trained for 60 epochs with AdamW, batch size 64, learning rate 2×10^{-3}, and weight decay 10^{-5}. The label-noise multiplier is $c = 0.05$. Code, checkpoints, split metadata, source CSV files, and plotting scripts are released with the paper.

7 Results

7.1 World-model fit and one-step coverage

The base FNO ends with training and validation MSEs 1.01×10^{-4} and 7.67×10^{-5}; the perturbed replica has validation MSE 9.87×10^{-5}. Under combined shift, the mean field L^2 error is 0.00780, and the perturbation scale has a modest radius–error correlation of 0.275. The scale therefore provides imperfect localization; calibration must absorb this misspecification.

Table 4 answers RQ1. The source-calibrated L^2 set falls to 0.741 field coverage. Deployment-audit calibration increases coverage, and the ellipsoid and simultaneous box both attain 0.906 on 800 independent combined-shift transitions. The box attains 1.000 decision-linear coverage but has more than twice the ellipsoid's mean support. Coverage is not tightness.

Table 4: Combined-shift one-step results. The target coverage is 0.90. Support is the half-width in the evaluated decision direction.

Set	Field coverage	Mean radius	Decision coverage	Mean support
Source L^2	0.741	0.01141	0.820	0.00650
Audit L^2	0.884	0.01882	0.934	0.01072
Audit ellipsoid	0.906	0.03242	0.986	0.02012
Audit simultaneous box	0.906	0.22729	1.000	0.04422

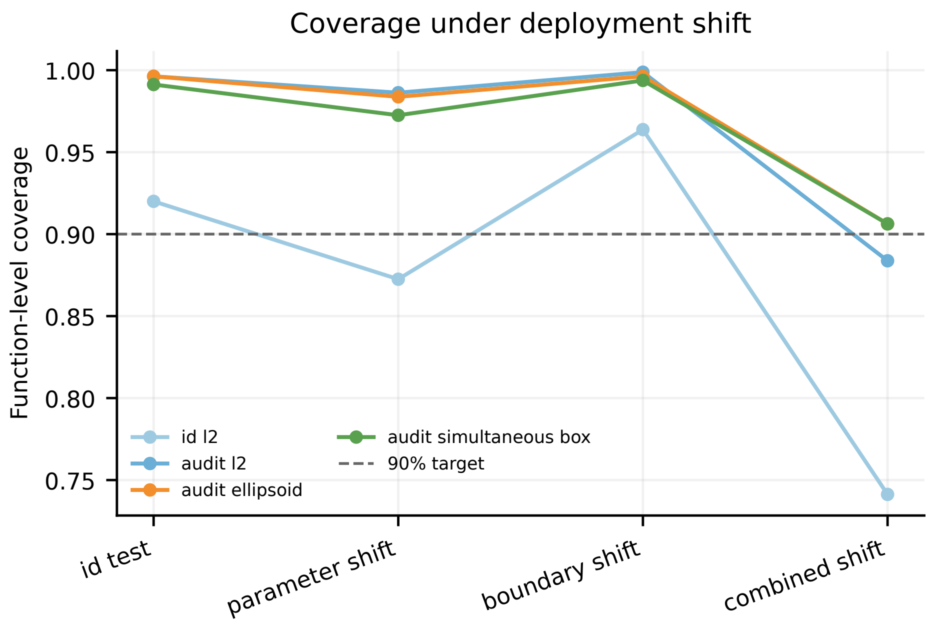

Figure 2: Realized one-step coverage across in-distribution, viscosity-shift, boundary-shift, and combined-shift test sets. Source calibration degrades most under combined shift; deployment-audit geometries recover the nominal region at different widths. Each regime contains 800 test transitions.

The separately calibrated behavior-trajectory max score attains empirical coverage 0.88 on 400 held-out length-10 trajectories at target 0.90. We report the observed proportion rather than rounding it to the target. Its scope is the fixed behavior-policy distribution in Equation (21).

7.2 Closed-loop control under latent actuator shift

Table 5 answers RQ2. Every robust controller reduces the empirical $p90$ relative to nominal MPC, but the ordering depends on geometry. The audit-box adjoint controller lowers mean cost by 1.89% and empirical $p90$ by 10.84%; audit L^2 lowers them by 1.92% and 7.79%. The ellipsoid adjoint and nonlinear adversarial variants improve less. Thus the result supports geometry-dependent decision benefit, not universal dominance of decision weighting or nonlinear inner maximization.

7.3 High-dimensional Navier–Stokes field audit

On the official 128×128 test set, the base FNO has mean relative L^2 error 0.1635. The max-type multiplier is $q_{0.90} = 30.679$; simultaneous test coverage is 0.883 and mean full coordinate-band width is 4.3409. The pointwise disagreement scale correlates only $r = 0.169$ with absolute error. This is a useful negative result: a valid score construction can remain inefficient when the maximum spans 16,384 coordinates and the local scale weakly ranks residual magnitude.

Table 5: Closed-loop cost over 24 matched actuator-gain cases. Changes are relative to nominal MPC; lower is better.

Controller	Mean	Mean change	$p90$	$p90$ change	Mean $\lvert a \rvert$
Nominal MPC	0.5797	–	1.0752	–	1.278
Source-L^2 robust	0.5724	-1.27%	1.0250	-4.66%	1.185
Audit-L^2 robust	0.5686	-1.92%	0.9914	-7.79%	1.115
Ellipsoid-adjoint	0.5747	-0.88%	1.0201	-5.12%	1.230
Box-adjoint	0.5688	-1.89%	0.9586	-10.84%	1.101
Adversarial ellipsoid	0.5764	-0.58%	1.0532	-2.04%	1.262

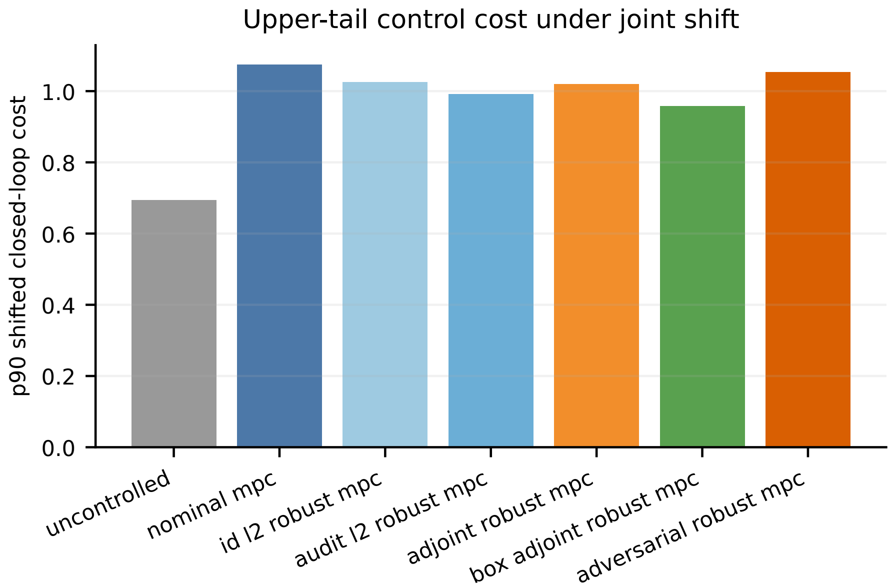

Figure 3: Empirical $p90$ closed-loop cost for the matched combined-shift actuator-gain sweep. The simultaneous-box adjoint controller gives the largest tail reduction, while the nonlinear adversarial controller is not best in this finite compute budget.

7.4 Value-gap scaling and theorem audit

RQ3 is tested independently of conformal calibration. For the scalar witness $G(x) = L_G x$, $\widehat{G}(x) = L_G x + \epsilon$, and reward $-L_r|x|$, the fixed-policy bound is attained. Across six values each of ϵ and γ, the log–log slope is 1.000 in ϵ. After normalizing the value gap by $\gamma\epsilon$, the slope against effective horizon $(1-\gamma)^{-1}$ is 2.000, with maximum relative numerical error 2.55×10^{-15}. The experiment verifies the theorem's quantity, not a conformal consequence.

For ten controlled Burgers rollouts with an exactly injected one-step error, the maximum observed gap-to-finite-horizon-bound ratio is 0.359. On the learned FNO's finite visited set, the maximum and mean gap-to-local-envelope ratios are 0.124 and 0.028. These are empirical local audits, not global PDE certificates.

7.5 Independent value-bound comparison

RQ5 compares four raw constructions after each is scaled on 60 calibration trajectories to the same 90% value-level target, then evaluated on 160 independent trajectories at horizon 20 and $\gamma = 0.95$. The local recursion has the smallest mean bound (0.0870) at 0.944 coverage. Adjoint support is wider (0.2035) at 0.925 coverage. The fitted curvature coefficient is zero, so the last two rows coincide. Decision weighting can improve a controller without producing the tightest scalar value certificate.

Table 6: Independent-test Navier–Stokes field-UQ results. No action channel is available, so no control metric is reported.

Relative L^2	$q_{0.90}$	Simultaneous coverage	Full width	Scale–error r
0.1635	30.679	0.883	4.3409	0.169

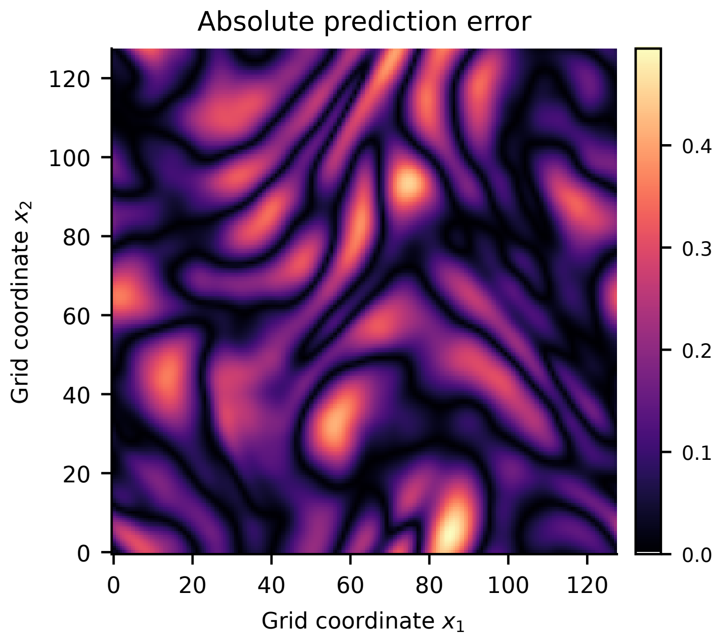

Figure 4: Representative absolute error of the two-dimensional FNO prediction. Errors are spatially structured rather than uniform, motivating a local scale; the following quantitative association test shows that the learned scale only weakly tracks them.

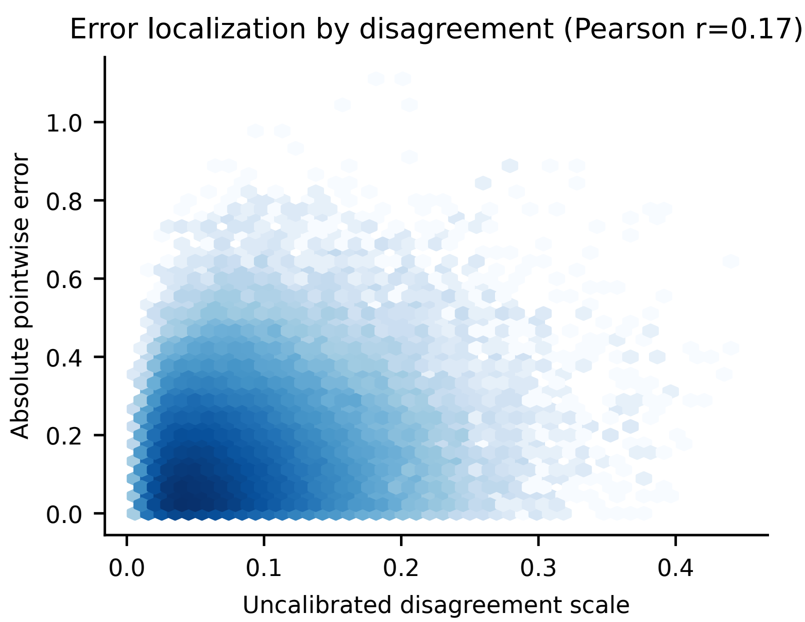

Figure 5: Pointwise perturbation scale versus absolute error on the independent Navier–Stokes test split. Pearson $r = 0.169$ reveals weak localization and helps explain the large simultaneous conformal multiplier.

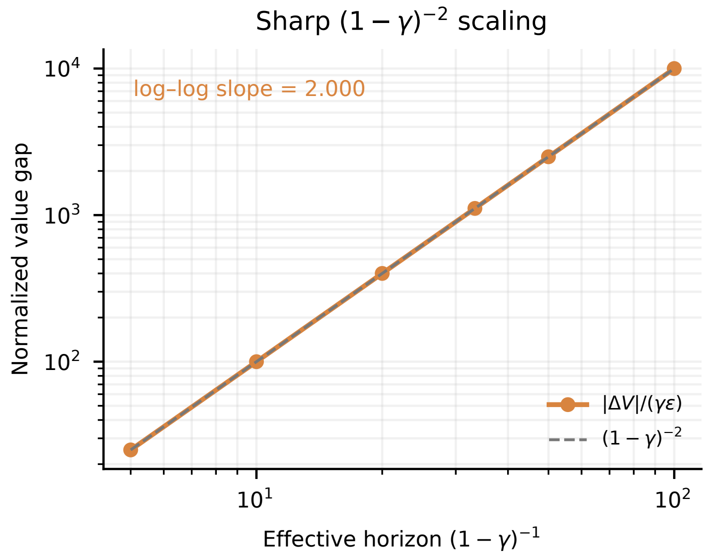

Figure 6: Sharp deterministic scaling audit. Normalizing by $\gamma\epsilon$ exposes the predicted quadratic dependence on the effective discount horizon. The generic policy-transfer bound is twice the fixed-policy witness plus optimization error.

Table 7: Independent-test bound comparison. Utilization is observed value gap divided by the reported bound. Coverage must be read together with width.

Method	Coverage	Mean bound	Median η	$p90$ η	Max η
Global maximum	0.938	0.0987	0.210	0.688	3.919
Local recursion	0.944	0.0870	0.322	0.788	2.190
Adjoint support	0.925	0.2035	0.265	0.854	1.766
Adjoint + curvature	0.925	0.2035	0.265	0.854	1.766

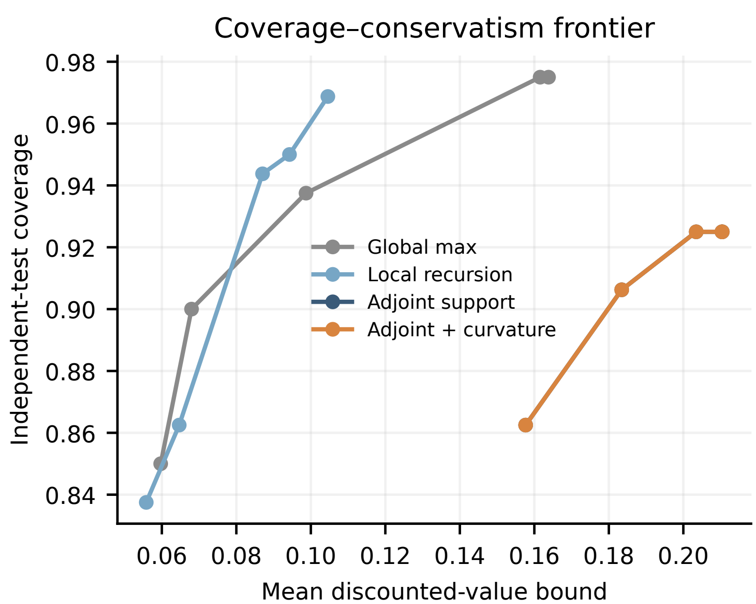

Figure 7: Coverage–mean-bound frontier on independent trajectories. A bound is less conservative only when it is narrower at matched coverage; utilization alone can be inflated by undercoverage.

8 Discussion and limitations

What the current evidence supports. Deployment-audit calibration repairs the source-set failure under compound shift, but different calibrated geometries lead to different support in the MPC adjoint direction. In the controlled actuator sweep, this distinction matters most in the upper tail: the simultaneous box is wide in field space yet gives the best empirical $p90$ controller. This finding illustrates the paper's central point: set width in a predictive norm and usefulness in a decision are not the same ordering.

What the current evidence does not support. The Burgers control result is one seed and one matched actuator sweep, not a multi-seed population claim. The public Navier–Stokes pairs have no action channel. The ordinary split-conformal theorem requires exchangeability with the deployment audit and does not cover a later distribution change. Marginal one-step coverage does not certify the product set queried by MPC, and the behavior-trajectory result does not cover counterfactual optimized actions. The deterministic value theorem assumes a uniform error and common closed-loop Lipschitz constants that were not certified for the learned FNO. The adjoint robust counterpart is first order unless a uniform curvature constant is established. Finally, finite CEM and PGD iterations introduce optimization errors that are measured empirically rather than bounded.

Negative results guide the next experiment. The nonlinear adversarial controller is not best; the adjoint scalar bound is not tighter than the local recursion; the curvature feature adds nothing; and the Navier–Stokes perturbation scale has weak error association. These outcomes argue for the following next tests: multiple training/control seeds; audit-size and shift-severity sweeps; learned low-rank covariance or spectral function-space sets; policy-coupled trajectory calibration; and a controlled two-dimensional PDE with an explicit action channel. Deep ensembles, probabilistic neural operators, and learned quantile operators should be added as scale baselines under a matched total label and compute budget [12], probabilistic neural operators [5], and calibrated quantile operators [15].

9 Conclusion

The value of this work is not a new neural-operator block. It is a falsifiable interface between prediction, uncertainty, and action. A perturbation-derived scale is first calibrated on a disjoint deployment audit to form a joint dynamics set over the discretized field. The set is then queried in the directions that change the finite-horizon objective, either analytically through exact support functions or numerically through adversarial rollouts. Finally, predictive coverage, deterministic value propagation, and closed-loop control are evaluated as separate claims.

The controlled-Burgers results show that this interface can improve empirical tail cost, but that the benefit is geometry-dependent and does not follow from coverage alone. The Navier–Stokes stress test shows the opposite pressure: simultaneous high-dimensional validity can demand wide sets when the local scale is weak. Together, these results sharpen the research question. The next advance should not merely produce a narrower prediction band; it should learn a function-space ambiguity geometry whose calibrated support is both valid for the deployment regime and useful for the decisions the controller actually makes.

References

[1] Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[2] Kavosh Asadi, Dipendra Misra, and Michael Littman. Lipschitz continuity in model-based reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 264–273, 2018.

[3] Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.

[4] Nicolas Boullé and Alex Townsend. A mathematical guide to operator learning. In *Handbook of Numerical Analysis*, volume 25, pages 83–125. Elsevier, 2024.

[5] Christoph Bülte, Philipp Scholl, and Gitta Kutyniok. Probabilistic neural operators for functional uncertainty quantification. *Transactions on Machine Learning Research*, 2025. arXiv:2502.12902.

[6] Kong Yao Chee, Thales C. Silva, M. Ani Hsieh, and George J. Pappas. Uncertainty quantification and robustification of model-based controllers using conformal prediction. In *Proceedings of the 6th Annual Learning for Dynamics and Control Conference*, volume 242 of *Proceedings of Machine Learning Research*, pages 528–540, 2024.

[7] Jose Leopoldo Contreras, Ola Shorinwa, and Mac Schwager. Safe, out-of-distribution-adaptive mpc with conformalized neural network ensembles. In *Proceedings of the 7th Annual Learning for Dynamics and Control Conference*, volume 283 of *Proceedings of Machine Learning Research*, pages 194–207, 2025.

[8] Anushri Dixit, Lars Lindemann, Skylar X. Wei, Matthew Cleaveland, George J. Pappas, and Joel W. Burdick. Adaptive conformal prediction for motion planning among dynamic agents. In *Proceedings of the 5th Annual Learning for Dynamics and Control Conference*, volume 211 of *Proceedings of Machine Learning Research*, pages 300–314, 2023.

[9] Amir-massoud Farahmand, Andre Barreto, and Daniel Nikovski. Value-aware loss function for model-based reinforcement learning. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 1486–1494, 2017.

[10] Chancellor Johnstone and Bruce Cox. Conformal uncertainty sets for robust optimization. In *Proceedings of the Tenth Symposium on Conformal and Probabilistic Prediction and Applications*, volume 152 of *Proceedings of Machine Learning Research*, pages 72–90, 2021.

[11] Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to PDEs. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

[12] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, volume 30, pages 6402–6413, 2017.

[13] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

[14] Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021.

[15] Ziqi Ma, Danielle Pitt, Kamyar Azizzadenesheli, and Anima Anandkumar. Calibrated uncertainty quantification for operator learning via conformal prediction. *Transactions on Machine Learning Research*, 2024.

[16] Emilia Magnani, Nicholas Krämer, Runa Eschenhagen, Lorenzo Rosasco, and Philipp Hennig. Approximate bayesian neural operators: Uncertainty quantification for parametric PDEs. *Transactions on Machine Learning Research*, 2025. arXiv:2208.01565.

[17] Emilia Magnani, Marvin Pförtner, Tobias Weber, and Philipp Hennig. Linearization turns neural operators into function-valued gaussian processes. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025.

[18] Ali Malik, Volodymyr Kuleshov, Jiaming Song, Danny Nemer, Harlan Seymour, and Stefano Ermon. Calibrated model-based deep reinforcement learning. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 4314–4323, 2019.

[19] David Millard, Lars Lindemann, and Ali Baheri. Split conformal prediction in the function space with neural operators, 2025.

[20] S. Chandra Mouli, Danielle C. Maddix, Shima Alizadeh, Gaurav Gupta, Andrew Stuart, Michael W. Mahoney, and Bernie Wang. Using uncertainty quantification to characterize and improve out-of-domain learning for pdes. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 36372–36418, 2024.

[21] Christian Moya, Ali Mollaali, Zhen Zhang, Lu Lu, and Guang Lin. Conformalized-DeepONet: A distribution-free framework for uncertainty quantification in deep operator networks. *Physica D: Nonlinear Phenomena*, 471:134418, 2025.

[22] NeuralOperator Team. Navier–stokes dataset, 2024.

[23] Anutam Srinivasan, Antoine Leeman, and Glen Chou. Safety beyond the training data: Robust out-of-distribution mpc via conformalized system level synthesis. In *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, volume 331 of *Proceedings of Machine Learning Research*, pages 412–439, 2026.

[24] Shyam Sundhar Ramesh, Pier Giuseppe Sessa, Yifan Hu, Andreas Krause, and Ilija Bogunovic. Distributionally robust model-based reinforcement learning with large state spaces. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 100–108, 2024.

[25] Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[26] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005.

[27] Weinan Wang, Bowen Gang, and Hao Deng. Operator learning for the 2d incompressible Navier–Stokes equations: A conformal prediction approach in the data-scarce regime, 2026.